Vibe-IC Canonical Flow v2.2.0

Plugin version: 0.2.2 · **Supersedes:** `CANONICAL_FLOW_v2.0.0.md`, `PHASE_VS_STAGE_VS_INDUSTRY_TAXONOMY.md`, `docs/tutorials/33_step_flow_overview.md`, `docs/design/STANDARD_FLOW.md` (all stop at the 33-step / 54-entity model and predate the 81-protocol Phase-1 dispatch, the Phase-2 scaffold bridge, and the doc→GDS LVS sign-off chain).

> **Source of truth = the runners, not this prose.** Every step below is cited to the actual step marker in its `*_one_shot_runner.py`. When a runner changes, regenerate this doc from the markers (`grep -noE '\[[0-9]+[a-z0-9]*/[0-9]+\]'` per runner) — do not hand-drift it.

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0. Entry paths + orchestrator

`programs/vibe_ic_one_shot_runner.py` is the top orchestrator. Two entry paths, one handoff:

```
Path A: NL prompt / dialogue ─┐
                              ├─► Phase 1 ─► generated_docs/L1-L23 JSON ─► Phase 2 ─► Phase 3
Path B: vendor design docs ───┘          (the ONLY universal handoff format)
```

Orchestrator sequence (`vibe_ic_one_shot_runner.py`): **Phase 1 → Phase 2 (=2a+2b) → Analog (after Phase 2, so L5_ADI_SPEC is populated; non-blocking) → Phase 3.** FAIL gating: Phase 1 FAIL halts before 2; Phase 2 FAIL halts before 3; Phase 3 FAIL → verdict FAIL but report emitted.

Phase	Transform	Runner	Gate
P0 pre-flight	env / PDK / tool availability	`mcp_server_health_check`, `eda_doctor`	tools reachable
Phase 1	any input → L1-L23 JSON (+ human MD)	`phase1_doc_one_shot_runner.py`	24 L-docs + completeness/ parity
Phase 2	L1-L23 → verified RTL → (gate netlist / FPGA SOF)	`phase2_one_shot_runner.py`	lint/synth/conformance/TB + final_audit
Analog A1-A8	L1/L5 → sized block → hardmacro (parallel to P2)	`analog_one_shot_runner.py`	per-block DRC/LVS + corners
Mixed-signal	digital + analog co-sim	skill `mixed-signal-cosim` (no dedicated runner)	M1-M4 co-sim
Phase 3	netlist → synth → PnR → GDS → DRC → LVS	`phase3_one_shot_runner.py`	DRC 0 + LVS + STA + tapeout checklist

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1. Phase 1 — 15 steps (`phase1_doc_one_shot_runner.py`, the `[N/15]` markers)

Marker	Step	Output
`[1/15]`	Extract text from `input/docs/` (PDF/ DOC/XLSX/...)	`input_doc/` (capped at 2 MB/doc for the O(n^2) scans, v0.1.91)
`[2/15]`	L1_DATASHEET	`generated_docs/ L1_DATASHEET.json`
`[3/15]`	L2_FRS	L2
`[4/15]`	L3_CMD_PROTOCOL	L3
`[5/15]`	L4_REGMAP	L4
`[6/15]`	L5_ADI_SPEC	L5
`[7/15]`	L6_CONTROL_LOGIC	L6

Marker	Step	Output
`[8/15]`	L7_TEST_DEBUG	L7
`[9/15]`	L8_RTL_CONSTANTS	L8_RTL_CONSTANTS
`[10/15]`	L9_INTEGRATION_SPEC	L9
`[11/15]`	L10_TEST_CASES	L10
`[12/15]`	L11_OTP_CONTENT	L11
`[13/15]`	L12_BEHAVIORAL_SEQUENCES	L12
`[14/15]`	L13_LAB_CALIBRATION	L13
`[14b/15]`	L8_TIMING_WAVEFORM (+ `14b2` width, `14b3` encoding, `14b7` universal constants, `14b4` L6 FSM, `14b5` L12 seq)	L8_TIMING_WAVEFORM + overlays
`[14c/15]`	L14-L18 protocol spec extract (+ `14c0` L9, `14c1` L1 meta, `14c1b` L17 handshake, `14c2` L3 mirror, `14c3` L17/ L18/L8_TIMING/L9 batch synth, `14c4` universal doc facts, `14c5` residual cleanup)	L14-L18
`[14d/15]`	L19-L23 skeleton emit	L19-L23
`[14e/15]`	**serial_peripheral_protocol class synth (R53/R54/R55)** — the protocol detector→synth dispatch	per-protocol L-doc overlays
`[14e2/15]`	bus_interconnect_protocol Tier-2 synth (TileLink/Wishbone/Avalon/OCP/AXI-Stream)	bus-protocol overlays
`[14e3/15]`	Universal packet/PDU L10↔L3 opcode-consistency sweep	L10 cleaned
`[15/15]`	Coverage / parity report	`reports/phase1/`

The `[14e/15]` block is the 81-protocol-class engine (this session, v0.1.84→v0.2.2). Each protocol ships a content-only `is_<proto>(blob)` detector + `<proto>_protocol_synth.py` overlay, dispatched by ic_class. Families: serial (SPI/I2C/I3C/UART/1-Wire/JTAG/SWD/QSPI/SMBus/MIPI-SPMI-RFFE), automotive (CAN/CAN-FD/LIN/ FlexRay/SENT/PSI5/Modbus/RS-485/CANopen), memory (DDR3/4/5/LPDDR5/HBM3/GDDR6/ONFI/ eMMC/SD-MMC/UFS/NVMe/HyperBus), display (HDMI/MIPI/DSI/CSI-2/DisplayPort/eDP), PCIe family (PCIe/Gen5/CXL/NVLink/ UCIe), USB (2.0/USB4), networking (Ethernet/800G/HDLC/SpaceWire/AFDX/Auto-Eth/ InfiniBand/FibreChannel/ PROFIBUS/PROFINET/IO-Link/EtherCAT), wireless (BLE/NFC/Zigbee/LoRa), audio (I2S/SoundWire/S-PDIF/A2B), data-converter (JESD204), debug (CoreSight), timing (PTP), aerospace (MIL-STD-1553/ARINC429), bus (AHB-APB/AXI-ACE-CHI/TileLink/Wishbone/Avalon/OCP/AXI-Stream), security (TPM). **Guard:** every module-level detector is auto-covered by `tests/test_protocol_detector_no_misfire.py` (no foreign-benchmark fire; derived-sibling allowlist in `protocol_detector_lib.DERIVED_SIBLING_CROSS_FIRES`).

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2. Phase 2 — RTL authoring + verification (`phase2_one_shot_runner.py`, `step_*`)

Order	Step (`def step_*`)	Notes
1	`step_phase1`	re-run/ingest Phase 1 if needed
2	`step_rig_topology_skeleton`	scaffold topology
3	`step_rtl_gen`	**WAIVES for ic_class with `rtl_gen=null`** → AI fills the `spec-to-rtl` role inside the pipeline (digital_arithmetic_primitive / digital_cmd_driven / serial_peripheral_protocol / bus_interconnect_protocol / processor_cpu / unknown). `phase2_scaffold_gen.py` emits top/

Order	Step (`def step_*`)	Notes
		regs/fsm/tb/soc_wrap/cocotb scaffold deterministically.
4	`step_full_stack_tb_gen`	self-checking TB generation
5	`step_reference_tb`	reference-TB conformance (eco_loop up to 3 retries)
6	`step_yosys_synth`	gate-level synth
7	`step_qsf_gen` / `step_sdc_gen`	FPGA project + constraints
8	`step_otp_image_check`	OTP image (if applicable)
9	`step_fpga_compile`	Quartus/yosys FPGA build → `.sof`
10	`step_fpga_burn`	program board (hardware path)
11	`step_usb_hid_tester_verify`	host-side protocol-tester acceptance
12	`step_emit_phase2_manifests`	phase2 manifests
13	`step_final_audit`	aggregate audit gate

Surrounding deterministic gates (fire around the AI authoring step): `rtl_hygiene_lint --fix`, `spec_conformance_check`, `chip_top_gate_wrapper_gen`, MCP `eda_lint`/`eda_synth`/`eda_cocotb`.

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3. Analog A1-A8 (`analog_one_shot_runner.py`, parallel to Phase 2)

Step	Name	Output
A1	spec_extract	`analog/<block>/A1_spec.json`
A2	topology_select	`A2_topology.json`
A3	netlist_gen	`<block>.sp`
A4	corner_sweep	`A4_corners.json`
A5	layout (Magic)	`A5_layout.json` (needs DRC-clean + LVS-match flags)
A6	post_layout_resim	`A6_postsim.json`
A7	hardmacro_gen	`{.lef,.lib,.gds,.v}` → feeds Phase 3
A8	hw_verify (HIL)	`A8_hw_verify.json`

(Older docs listed "A1-A9"; the runner ships **A1-A8**. Mixed-signal **M1-M4** is skill-level `mixed-signal-cosim`, no dedicated `*_one_shot_runner.py`.)

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4. Phase 3 — physical design + sign-off (`phase3_one_shot_runner.py`, `step_*`)

Step	Name	Tool (open-source substitute)
1	`step_synth`	yosys (sky130/gf180) — **+ tie-cell pass** (see § 5)
2	`step_pnr`	OpenROAD (floorplan → PDN → place → CTS → route)
3	`step_gds`	KLayout streamout (`def2gds`; OpenROAD no longer streams GDS)
4	`step_drc`	KLayout sky130 deck
5	`step_lvs`	netgen / yosys_equiv — **the sign-off chain in § 5**
6	`step_canonicalize_artefacts`	normalize outputs

Tapeout gate: `tapeout-checklist` (DRC/LVS/STA/IR/EM/antenna/ERC/LEC/DFT) + `flow_compliance_check`.

5. The LVS sign-off chain — NEW in this era (v0.1.96 → v0.2.2)

This is the part **entirely missing from the old docs**. `step_lvs` is no longer a single yosys_equiv call; it is a layered chain whose every layer was hardened by the 7 doc→GDS pilots:

1. **Structural LEC (default)** — `eda_lvs mode=yosys_equiv` (`equiv_simple`+`equiv_induct`). Residual "unproven" cells = yosys SAT-model gap on PDK primitives (Category-D tool limit, NOT a mismatch). No yosys flag closes it for all cells.
2. **Device-level coverage** — to cover the SAT gap: `eda_extraction` (magic ext2spice) + `eda_lvs mode=netgen` + `lvs_netgen_setup_emit.py` (power-net globalization). Compares transistors → no SAT-model concept. Reaches device-class-exact (HDLC 20937=20937; sha256 12148=12148).
3. **Powered-netlist closure** — OpenROAD `write_verilog -include_pwr_gnd` (after `global_connect`) gives the schematic side real VPWR/VGND/VPB/VNB → eliminates the tie-cell disconnected-node residual.
4. **Top-level port labels** — Route A (canonical): `magic_port_extract_emit.py` (`export PDK` + `port makeall`). Route B (fallback): `lvs_def_port_seed.py` (parse DEF PINS). (Route A is required; B is an audit hint — empirically confirmed.)
5. **Sign-off guard (MANDATORY)** — `lvs_signoff_guard.py`: RAISES on a claimed match against a PORTLESS extracted top `.subckt` (the vacuous / silent-false-positive condition). Run before trusting ANY LVS match.

Synth tie-cell pre-step (§ 4 step 1): `setundef -zero; hilomap -hicell conb_1 HI -locell conb_1 LO; splitnets; clean` (NO `opt_clean` — it deletes tie cells). Without it, constant nets fail TritonRoute DRT-0305. The bare MCP `eda_synth` path lacks this (backlog `ORGANIC-20260531-mcp-eda-synth-missing-hilomap-tiecells`); `phase3_one_shot_runner` does it automatically. Forward-validated SENT→QSPI→HDLC→SpaceWire.

6. doc→GDS pilot evidence (7 pilots, real sky130A GDS)

Pilot	Archetype	ic_class	LVS stop point
i2s	streaming-rx	digital_cmd_driven	structural (3 SAT-unproven)
ahb_apb	bus-bridge	bus_interconnect	—
ufs	storage-framer	serial_peripheral	—
sent	sensor-decoder	digital_arithmetic_primitive	structural all-proven 1388/1388
qspi	command-controller	serial_peripheral	structural all-proven 1434/1434
hdlc	packet-framer	digital_cmd_driven	**device-level exact 20937=20937**
spacewire	link credit-flow-control	digital_arithmetic_primitive	structural 493/592 (device-level closes the SAT gap; not yet re-run)

7. Project folder layout (unchanged from v2.0.0)

```
<project>/
  input/{docs,phase1_prompt.md}      input_doc/           (Phase 1 in/extracted)
  phase1/{generated_docs,human_docs,claude_extracted}/L*.json|md
  phase2/stage1/{rtl,scaffold,fpga}/  phase2/stage2/synth/
  analog/<block>/{A1_spec.json,…,<block>.{sp,lef,lib,gds,v}}
  phase3/stage{1..4}/{synth,pnr,gds,drc,lvs}/  phase3/stage5_manufacturing/
  reports/{phase1,phase2,phase3,orchestrator}/
```

8. Keeping this current

This doc is **derived**, not authoritative. To regenerate after a runner change: `grep -noE '\[[0-9]+[a-z0-9]*/[0-9]+\][^"]*' phase1_doc_one_shot_runner.py` and `grep -noE 'def step_[a-z0-9_]+' phase{2,3}_one_shot_runner.py` and `analog_one_shot_runner.py`'s header A1-A8 block. A future enhancement (backlog candidate): a `programs/flow_doc_emit.py` that emits this table deterministically from the runner markers, the way `INDEX.md` is generated.